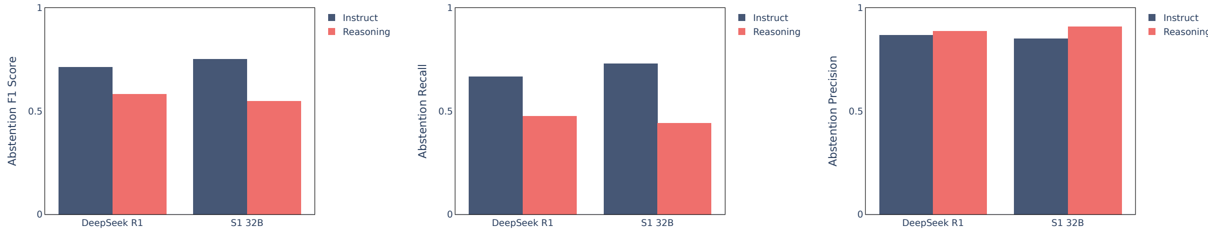

Fig. S9 Comparing the effect of reasoning across all datasets. We pair each reasoning model with its instruction tuned model: DeepSeek R1 v. Llama 3.3 70B Instruct and S1.1 32B v. Qwen 2.5 32B.

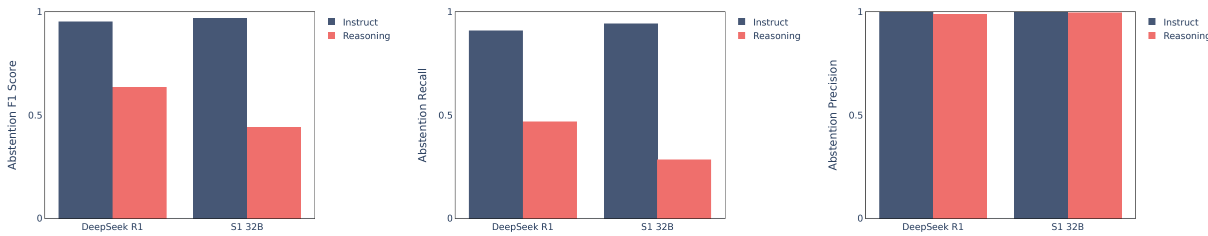

Fig. S10 Comparing the effect of reasoning on reasoning datasets. We pair each reasoning model with its instruction tuned model: DeepSeek R1 v. Llama 3.3 70B Instruct and S1.1 32B v. Qwen 2.5 32B.

the standard chat formatting.[2] To generate a reasoning chain for s1.1 and DeepSeek, we allocate max of $4k$ tokens and set stop tokens in vLLM inference to the end-of-reasoning tokens (`</think>` in DeepSeek and `<|im_start|>answer` for s1.1).

To generate the final answer, we concatenate the formatted original prompt with the generated reasoning chain and end-of-reasoning tokens and allocate another $4k$ max tokens for the answer. Following Muennighoff et al. [14], in addition to the end-of-reasoning tokens we add a `\nFinal Answer:` string for s1.1 and a `\n\n**Final Answer**\n\\boxed{` string for DeepSeek R1 Distill to interrupt reasoning and generate the final answer. We find this to be a helpful approach to control the generation of the final answer in s1.1, however, for DeepSeek we find that on some datasets the model ignores the end-of-reasoning tokens and continues the reasoning chain generation. Recent work [92] has also also reported challenges with force terminating DeepSeek R1 Distill's reasoning chain, with the model ultimately generating multiple `</think>` tokens in its response.

When evaluating s1.1 on reasoning datasets, we follow Muennighoff et al. [14] and use greedy decoding (temperature $= 0$) and find that it has a significant positive effective on accuracy.

Reasoning model results. We show the effect of reasoning on recall, precision and F1 score across all datasets in Fig. S9 and on reasoning datasets only in Fig. S10.

DeepSeek R1 vs Llama 3.3. In Fig. S11 we compare Llama 3.3 70B Instruct and DeepSeek R1 Distill Llama (which is a distillation from the full DeepSeek R1 into a Llama 3.3 model). We find that reasoning fine-tuning on Llama 3.3 significantly improves accuracy on GPQA-Abstain while resulting in minor accuracy improvements or degradations on the other reasoning datasets. We note that reasoning fine-tuning improves accuracy on the two reasoning datasets which have multiple-choice question formats, while slightly degrading performance on open-ended questions from GSM8K-Abstain and UMWP. We also note that Llama 3.3 accuracy on both GSM8K-Abstain and UMWP is above 97%. However, across all reasoning datasets reasoning fine-tuning significantly harms abstention recall.

[2] See https://github.com/simplescaling/s1 for an example of the same approach to forced-reasoning formatting by the authors of s1.

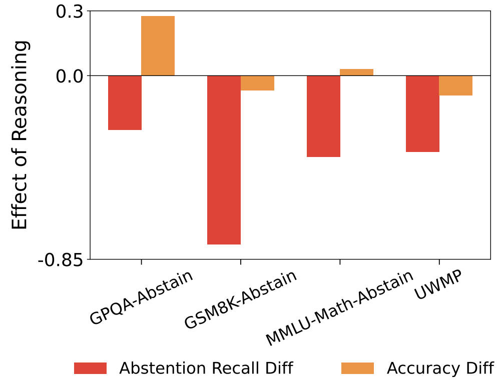

Fig. S11 Comparison of DeepSeek R1 Distill and Llama 3.3 70B in terms of abstention recall and accuracy on reasoning datasets.

Reasoning budget on DeepSeek and s1. In Fig. 7, we show the effect of reasoning budget scaling for s1.1 when using greedy decoding, as recommended in Muennighoff et al. [14]. In Fig. S12 we show a similar trend preserves when using temperature 0.8 (used for inference in other LLMs), however, the accuracy on answerable questions is generally higher when using temperature 0.

In Fig. S13 we show the effect of reasoning budget on DeepSeek R1 Distill. We use the same values for max reasoning token as in s1.1: {512, 768, 1024, 2048, 4096}. We see that on GSM8K-Abstain and UMWP the average number of reasoning tokens is actually much smaller than the set maximum – the model exits the reasoning much earlier, often below 512 tokens. On GPQA-Abstain and MMLU-Math-Abstain we see a similar trend to s1.1, with higher reasoning budget leading to generally better accuracy and lower abstention recall.

We note that results on Fig. S13 were generated when using an additional "trigger" token for interrupting the reasoning chain and starting the final answer generation. In Fig. S14 we show analogous results when not using this trigger token. We can see that the empirical average number of tokens used for reasoning is quite different from our set maximum, and it is more challenging to steer DeepSeek R1 to stop "thinking" on GPQA-Abstain and MMLU-Math.

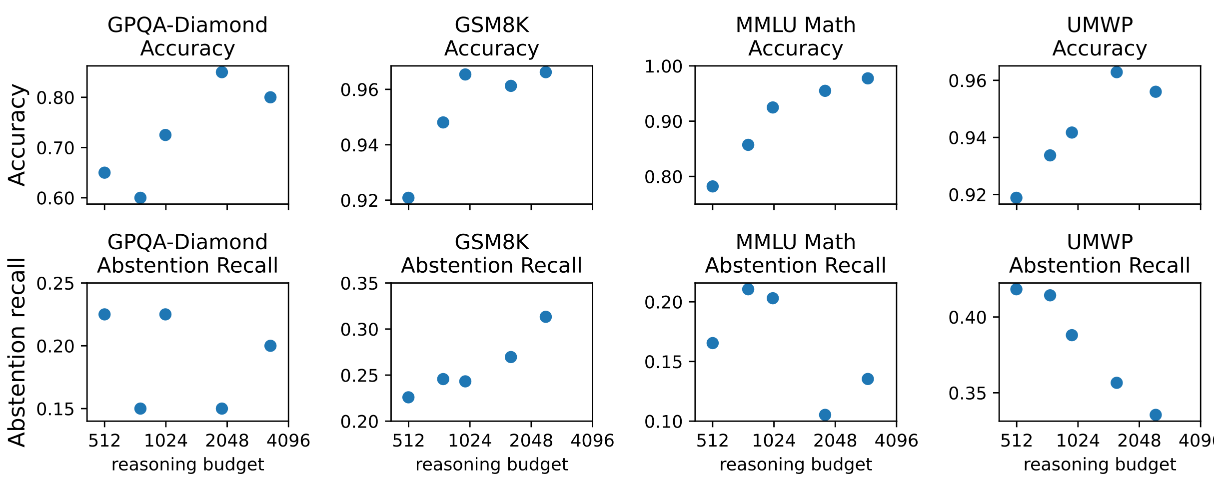

Fig. S12 s1.1 test-time scaling via reasoning token budget for inference with temperature 0.8.

o1 reasoning effort. We show the effect of adjusting the reasoning effort parameter exposed by OpenAI in Table 5, which shows the default reasoning effort leads to the best abstention performance, though we have no way of knowing what effect this parameter has in the pipeline behind the API.

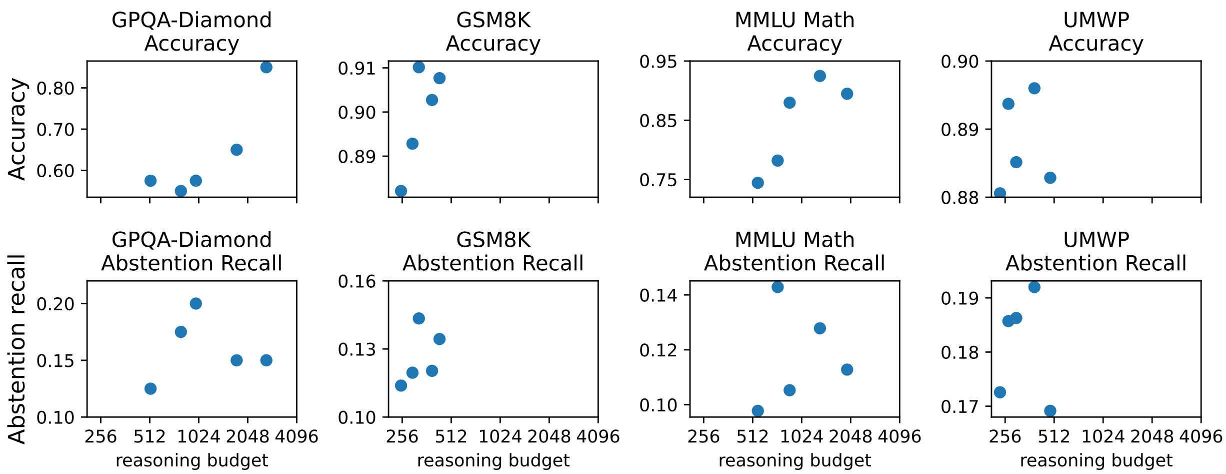

Fig. S13 DeepSeek R1 Distill 70B test-time scaling via increasing reasoning token budget on reasoning datasets. When forcing final answer generation, we append an additional string which triggers DeepSeek to stop the reasoning chain with higher rate.

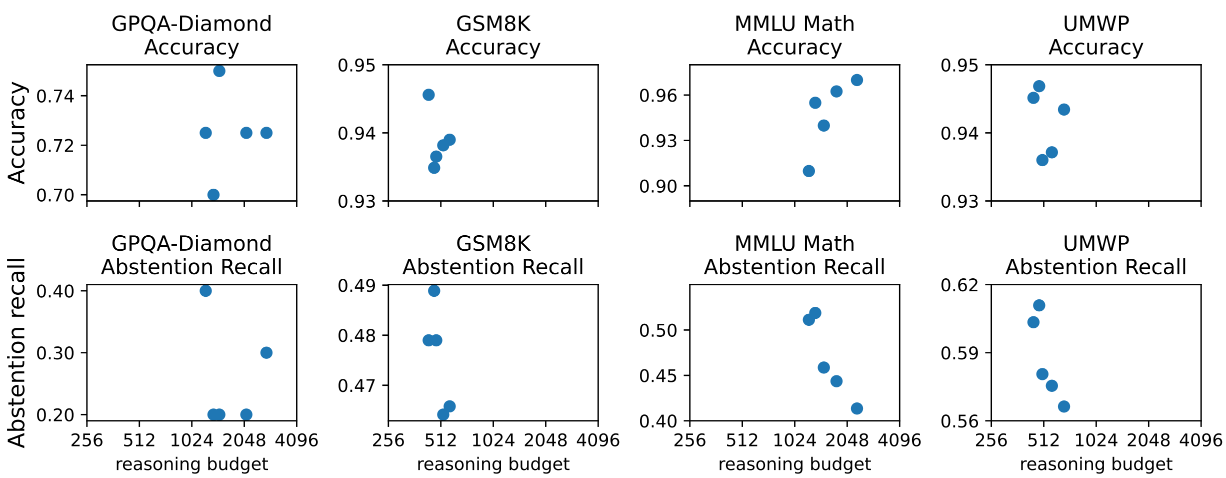

Fig. S14 DeepSeek R1 Distill 70B test-time scaling via increasing reasoning token budget on reasoning datasets.

Table 5 Abstention recall on reasoning datasets for various settings of reasoning effort in o1 API model.

reasoning_effort	GPQA-Diamond	GSM8K	MMLU-Math
low	0.63	0.96	**0.71**
medium (default)	**0.78**	**0.95**	0.70
high	0.60	0.95	0.68

LLM judge which evaluates both reasoning chain and final answer. In Fig. S15 we show abstention precision for DeepSeek R1 and s1.1 when using a regular abstention LLM judge which only relies on the final answer for detecting abstention versus the judge which uses both verbose reasoning chain and final answer for scoring abstention. We can see that the reasoning chains are more likely to contain expressions of uncertainty which leads to higher recall and lower precision when using reasoning chain in evaluation. Even when the models provide caveats in their reasoning chains, they still often generate a confident final answer.